



Home



Course



Games



Discussions



Lessons



Flashcards



Practice Questions



Tests



Search

< >

Division

Table of Contents

Confidence Levels

Notes

Bookmarks

Highlights

Toolbox



Division



Division

Long division is a four-step process that repeats until there are no digits left to bring down. Any nonzero number left over after the final step is called a remainder. The remainder must always be less than the number you divided by.

Four-Step Division Process

Step 1. Divide and write the answer as part of the quotient.

Step 2. Multiply the answer by the divisor.

Step 3. Subtract.

Step 4. Bring down the next digit and repeat the process if needed.

Example

Hide

Frank needs to package 426 designer cupcakes. If a box holds 8 cupcakes, how many boxes can he fill?

Estimate: $400 \div 8 = 50$

Step 1

Divide 8 into 42. Write the answer above the 2.

divisor ← $\overline{8)426}$ → dividend

5 → quotient

Step 2

Multiply: $5 \times 8 = 40$

$$\begin{array}{r} 5 \\ 8 \overline{)426} \\ \underline{-40} \end{array}$$

$(5 \times 8 = 40)$

Step 3

Subtract: $42 - 40 = 2$

$$\begin{array}{r} 5 \\ 8 \overline{)426} \\ \underline{-40} \\ 26 \end{array}$$

Step 4

Bring down the next digit. Repeat.

$$\begin{array}{r} 53R2 \\ 8 \overline{)426} \\ \underline{-40} \\ 26 \\ \underline{-24} \\ 2 \end{array}$$

$(3 \times 8 = 24)$

The remainder is 2. Write the remainder after the letter R.

Frank can fill 53 boxes. He will have 2 cupcakes left over.

Check: Your answer of **53** is reasonably close to 50.

Example 2:

Estimate:

$$\begin{array}{r} 92 \\ 7 \overline{)645} \\ \underline{63} \downarrow \\ 15 \\ \underline{14} \\ 1 \\ R=1 \end{array}$$

$$\begin{array}{r} 630 \\ 7 \\ \hline = 90 \end{array}$$

so $645 \div 7 = 92,$

with 1 left over.

Check: $90 \times 7 = 63$
 $2 \times 7 = 14$
 $\underline{63} + 14 = 77$
 $77 + 1 = 78$

Rate Your Confidence

^ High

= Medium

v Low

Continue >>

Category

Lesson 2: Operations with Whole Numbers and Decimals

Related Questions:

[Practice questions taken related to this topic](#)

Report Content Errors

